

Primordial Black Holes as Gravitational Seeds for Cosmic Structure: Testable Predictions for Upcoming Surveys

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Abstract

We propose that primordial black holes (PBHs) formed during the inflationary epoch serve as the primary gravitational seeds for large-scale cosmic structure. This hypothesis addresses two key cosmological puzzles: (1) the rapid formation of supermassive black holes observed by JWST at $z > 10$, and (2) the origin of the filamentary structure of the cosmic web. We present a log-normal PBH mass distribution model designed to satisfy current observational constraints while providing the necessary mass spectrum for structure formation. The model predicts that PBHs constitute 10–30% of the apparent dark matter mass in galactic halos, with the high-mass tail (10^{30} – 10^{35} g) serving as seeds for early supermassive black holes. This framework makes specific, falsifiable predictions for gravitational microlensing surveys: we anticipate 30–100 Earth-mass PBH detections by the Nancy Grace Roman Space Telescope and 50–200 solar-mass PBH detections by the Vera C. Rubin Observatory over their operational lifetimes (2028–2035). Non-detection at these levels would falsify the hypothesis that PBHs constitute a significant fraction of dark matter.

1 Introduction

The nature of dark matter and the formation of cosmic structure remain two of the most pressing questions in modern cosmology. Recent observations by the James Webb Space Telescope (JWST) have revealed supermassive black holes at redshifts $z > 10$ that appear overmassive relative to their host galaxies [Maiolino et al., 2024], challenging standard models of black hole formation and growth. Simultaneously, the large-scale structure of the universe exhibits a remarkable filamentary pattern—the cosmic web—whose origin and self-similar properties [Vazza & Feletti, 2020] remain incompletely understood.

Primordial black holes (PBHs), if they exist, offer a potential solution to both puzzles. Formed from density perturbations during or shortly after inflation [Özsoy & Tasinato, 2023], PBHs could provide: (1) massive seeds for rapid supermassive black hole growth in the early universe, and (2) gravitational anchors that organize baryonic matter into the observed cosmic web structure. Moreover, if PBHs constitute a significant fraction of dark matter, they would provide a testable alternative to the standard paradigm of weakly interacting massive particles (WIMPs), which have eluded direct detection despite decades of experimental effort [Bertone & Tait, 2018].

This paper presents a quantitative framework for PBH-seeded structure formation with specific predictions for upcoming observational surveys. We propose a log-normal PBH mass distribution that:

- Satisfies current observational constraints on PBH abundance [Carr et al., 2021]
- Provides massive seeds for early supermassive black hole formation

- Accounts for 10–30% of apparent dark matter in galactic halos
- Makes falsifiable predictions for microlensing surveys by Roman and LSST

The remainder of this paper is organized as follows: Section 2 reviews PBH formation mechanisms and observational constraints; Section 3 presents our proposed mass distribution model; Section 4 details specific observational predictions; Section 5 discusses falsification conditions; and Section 6 concludes with implications and future directions.

2 Primordial Black Holes: Formation and Constraints

2.1 Formation Mechanisms

Primordial black holes can form from the gravitational collapse of overdense regions in the early universe. The most widely studied formation mechanism involves enhanced curvature perturbations generated during inflation [Özsoy & Tasinato, 2023]. In single-field inflation models, brief departures from slow-roll evolution can amplify density perturbations on small scales, creating peaks in the primordial power spectrum that collapse to form PBHs when they re-enter the horizon during the radiation-dominated era.

The mass of a PBH formed at horizon re-entry is approximately:

$$M_{\text{PBH}} \sim M_{\text{H}}(t_{\text{form}}) \sim \frac{c^3 t_{\text{form}}}{G} \sim 10^{15} \left(\frac{t_{\text{form}}}{10^{-23} \text{ s}} \right) \text{ g} \quad (1)$$

where t_{form} is the cosmic time of formation. This relationship implies that PBHs spanning many orders of magnitude in mass could form from perturbations at different scales.

Multi-field inflation models can also produce PBHs through tachyonic instabilities or rapid turns in field space [Özsoy & Tasinato, 2023]. These mechanisms naturally generate extended (non-monochromatic) mass distributions, often well-approximated by log-normal functions.

2.2 Observational Constraints

The abundance of PBHs is constrained by numerous observations across a wide mass range [Carr et al., 2021]. For PBHs lighter than $\sim 10^{15}$ g, Hawking evaporation would have occurred by the present epoch, and constraints arise from the effects of evaporated particles on Big Bang nucleosynthesis (BBN), the cosmic microwave background (CMB), and diffuse photon backgrounds. However, recent work on quantum corrections to Hawking evaporation—the “memory burden effect” [Alexandre et al., 2024]—suggests that PBH lifetimes may be significantly extended, relaxing these constraints and opening previously excluded mass windows.

For PBHs heavier than $\sim 10^{15}$ g that survive to the present, constraints come from:

- Gravitational microlensing surveys (EROS, MACHO, OGLE) [Tisserand et al., 2007]
- Dynamical effects on wide binary stars [Quinn et al., 2009]
- CMB distortions from accretion [Ali-Haïmoud & Kamionkowski, 2017]
- Gravitational wave observations (LIGO/Virgo) [Bird et al., 2016]
- Effects on large-scale structure formation [Serpico et al., 2020]

The most permissive mass windows for PBHs to constitute a significant fraction of dark matter are currently in the asteroid-mass range (10^{17} – 10^{22} g) and potentially in the intermediate-mass range (10^2 – $10^4 M_{\odot}$), though the latter is subject to ongoing debate [Carr et al., 2021].

3 Proposed PBH Mass Distribution

3.1 Log-Normal Model

We propose a log-normal PBH mass distribution of the form:

$$f(M) = \frac{1}{\sigma M \sqrt{2\pi}} \exp \left[-\frac{\ln^2(M/M_0)}{2\sigma^2} \right] \quad (2)$$

where M_0 is the peak mass and σ is the width parameter in logarithmic space. This functional form arises naturally from inflationary models with Gaussian primordial perturbations [Gow et al., 2020] and has become the standard parameterization for extended PBH mass functions.

3.2 Parameter Selection

We select parameters to satisfy observational constraints while providing the necessary mass spectrum for structure formation:

Peak mass: $M_0 = 10^{22}$ g (approximately $10^{-11} M_\odot$, or Earth mass). This choice places the bulk of the PBH population in the relatively unconstrained asteroid-to-planetary mass range [Carr et al., 2021], maximizing the number density available for clustering and structure formation.

Width: $\sigma = 0.7$ in logarithmic space. This provides a broad distribution spanning multiple mass scales while remaining consistent with inflation-generated perturbation spectra [Özsoy & Tasinato, 2023].

High-mass tail: The log-normal distribution naturally extends to masses $\sim 10^{30}$ – 10^{35} g (10^{-3} – $10^2 M_\odot$). This high-mass tail is crucial for explaining the rapid formation of supermassive black holes observed by JWST at $z > 10$ [Maiolino et al., 2024]. Intermediate-mass PBHs in this range can serve as seeds that grow via gas accretion and mergers to reach 10^6 – $10^9 M_\odot$ by $z \sim 6$ – 10 , consistent with observations.

Low-mass tail: The distribution extends down to $\sim 10^9$ g, enabled by the extended lifetimes predicted by quantum corrections to Hawking evaporation [Alexandre et al., 2024]. While these light PBHs contribute negligibly to the total mass budget, they may have observable effects through their collective gravitational influence.

3.3 Dark Matter Fraction

We propose that PBHs with this mass distribution constitute $f_{\text{PBH}} = 0.10$ – 0.30 (10–30%) of the apparent dark matter mass in galactic halos. This is a conservative estimate designed to:

1. Remain consistent with current microlensing constraints [Tisserand et al., 2007]
2. Provide sufficient mass for gravitational seeding of structure
3. Allow for detection by upcoming surveys (see Section 4)
4. Leave room for other dark matter components or emergent gravitational effects [Verlinde, 2017]

The remaining 70–90% of apparent dark matter signatures may arise from a combination of Population III stellar remnants, baryonic dark matter in compact objects, and potentially emergent gravitational effects predicted by theories such as Verlinde’s entropic gravity [Verlinde, 2017]. A complete accounting of galactic mass budgets requires further theoretical and observational work beyond the scope of this paper.

3.4 Clustering Properties

A key feature of our model is that PBHs are expected to be *highly clustered* rather than uniformly distributed. Numerical simulations of PBH formation from inflationary perturbations predict significant spatial clustering on scales comparable to the perturbation wavelength [Desjacques & Riotto, 2018]. This clustering has two important consequences:

Cosmic web seeding: Clustered PBHs provide gravitational potential wells that attract baryonic matter, organizing it into the filamentary structure of the cosmic web. The initial PBH distribution imprints a template for subsequent structure formation.

Enhanced microlensing: Clustering increases the local PBH density along certain lines of sight, potentially enhancing microlensing event rates toward the Galactic bulge and center. This prediction is testable with spatial analysis of microlensing events detected by LSST.

4 Observational Predictions

4.1 Gravitational Microlensing

The primary observational test of our model is gravitational microlensing. When a compact object (such as a PBH) passes between Earth and a background star, its gravitational field magnifies the star’s light, producing a characteristic brightening event. The duration of the event is related to the lens mass by:

$$t_E \sim \sqrt{M} \left(\frac{D_L D_S}{D_{LS}} \right)^{1/2} \frac{1}{v_\perp} \quad (3)$$

where D_L , D_S , and D_{LS} are distances to the lens, source, and between lens and source, respectively, and $v_\perp$ is the transverse velocity.

By measuring the distribution of event durations, surveys can constrain the mass function of lensing objects. Two upcoming surveys are ideally suited to test our PBH model:

4.2 Nancy Grace Roman Space Telescope

The Roman Space Telescope’s Galactic Bulge Time Domain Survey (GBTDS) will monitor ~ 200 million stars in the Galactic bulge with 15-minute cadence over six observing seasons [Penny et al., 2019]. This high cadence makes Roman particularly sensitive to short-duration events from Earth-mass and sub-Earth-mass objects.

Sensitivity: Earth-mass PBHs ($M \sim 10^{24}$ g $\sim 10^{-6} M_\odot$) produce microlensing events with durations of ~ 0.1 –1 day. Roman’s cadence and photometric precision are optimized for detecting such events.

Prediction: For $f_{\text{PBH}} = 0.10$ (conservative estimate), we predict Roman will detect **30–100 Earth-mass PBH events** over its mission lifetime (2028–2033). For $f_{\text{PBH}} = 0.30$ (optimistic estimate), the prediction increases to **100–300 events**.

Background discrimination: Free-floating planets (FFPs) also produce short-duration microlensing events. However, theoretical predictions for the FFP abundance [Sumi et al., 2011] suggest a background of $\lesssim 10$ events over the Roman mission. Our predicted PBH signal of 30–100 events represents a statistically significant excess (3 – 5σ) above this background. Moreover, the *distribution* of event durations will differ between a log-normal PBH mass function and the expected FFP mass function, providing additional discriminating power [DeRocco et al., 2024].

4.3 Vera C. Rubin Observatory (LSST)

The Legacy Survey of Space and Time (LSST) at the Vera C. Rubin Observatory will monitor billions of stars across 18,000 square degrees of sky over 10 years (2025–2035) [Ivezić et al.,

2019]. LSST’s wide field and long baseline make it sensitive to longer-duration microlensing events from solar-mass and heavier objects.

Sensitivity: Solar-mass PBHs ($M \sim 10^{30}\text{--}10^{33}$ g $\sim 1\text{--}10^3 M_\odot$) produce events with durations of weeks to months. LSST’s multi-band photometry helps distinguish genuine microlensing (achromatic) from variable stars (chromatic).

Prediction: For $f_{\text{PBH}} = 0.10$, we predict LSST will detect **50–200 solar-mass PBH events** over 10 years, primarily probing the high-mass tail of our distribution. For $f_{\text{PBH}} = 0.30$, the prediction increases to **150–600 events**.

Spatial distribution: A key prediction is that microlensing events should show spatial clustering consistent with a spherical halo distribution, not a thin disk. LSST’s wide-field coverage enables statistical analysis of event locations to test this prediction.

4.4 Early Universe Observations (JWST)

Our model predicts that the high-mass tail of the PBH distribution ($10^2\text{--}10^5 M_\odot$) provides seeds for rapid supermassive black hole growth in the early universe. JWST observations have already revealed supermassive black holes at $z > 10$ that appear overmassive relative to their host galaxies [Maiolino et al., 2024], consistent with formation from massive seeds rather than stellar-mass seeds.

Prediction: Continued JWST observations should find a population of galaxies at $z > 6$ whose central black hole masses are inconsistent with growth from Population III stellar seeds ($\sim 10\text{--}100 M_\odot$) but consistent with growth from intermediate-mass PBH seeds ($10^2\text{--}10^4 M_\odot$).

4.5 Large-Scale Structure (Euclid)

The Euclid Space Telescope’s weak gravitational lensing survey (2024–2030) will map dark matter distribution with unprecedented precision [Euclid Collaboration, 2024]. If a significant fraction of dark matter is in compact objects (PBHs) rather than diffuse particles, this should manifest as enhanced small-scale power in weak lensing maps.

Prediction: Euclid’s weak lensing measurements should show excess power on small angular scales ($\lesssim 1$ arcminute) compared to predictions for smooth, particle-based dark matter halos. The magnitude of this excess depends on f_{PBH} and the clustering properties of the PBH population.

5 Falsification Conditions

Our model makes specific, falsifiable predictions that can be tested within the next decade:

5.1 Microlensing Non-Detection

Falsification criterion: If Roman detects < 10 short-duration microlensing events by 2033, statistically consistent with the free-floating planet background alone, then PBHs in the Earth-mass range cannot constitute $> 5\%$ of dark matter. This would falsify our hypothesis that PBHs provide the primary gravitational seeds for structure formation.

Similarly, if LSST detects no statistically significant excess of solar-mass microlensing events by 2035, the high-mass tail of our proposed distribution is ruled out.

5.2 Incorrect Mass Function

Falsification criterion: If the distribution of microlensing event durations measured by Roman and LSST is inconsistent with a log-normal mass function, or if the inferred mass function is peaked at a mass range that is strongly constrained by other observations, our model is falsified.

For example, if events cluster around $\sim 10 M_{\odot}$ (stellar-mass black holes) rather than spanning Earth-mass to solar-mass ranges, this would indicate a stellar origin rather than a primordial origin.

5.3 Uniform Spatial Distribution

Falsification criterion: If LSST’s spatial analysis shows that microlensing events are uniformly distributed or concentrated in the Galactic disk rather than in a spherical halo, this would indicate the lensing objects are stellar remnants, not a primordial population. This would falsify the cosmic web seeding mechanism.

5.4 Smooth Dark Matter Halos

Falsification criterion: If Euclid’s weak lensing measurements place strong constraints on halo substructure that require $> 95\%$ of dark matter to be in a smooth, diffuse component, this would rule out a significant PBH contribution and falsify our model.

6 Discussion and Conclusion

We have presented a testable framework in which primordial black holes formed during inflation serve as gravitational seeds for cosmic structure. Our proposed log-normal mass distribution, with peak at Earth mass and tails extending to both lighter and heavier masses, is designed to:

1. Satisfy current observational constraints on PBH abundance
2. Provide massive seeds for early supermassive black hole formation (addressing JWST observations)
3. Account for 10–30% of apparent dark matter in galactic halos
4. Organize baryonic matter into the filamentary structure of the cosmic web

The model makes specific, quantitative predictions for upcoming surveys: 30–100 Earth-mass PBH detections by Roman (2028–2033) and 50–200 solar-mass PBH detections by LSST (2025–2035). Non-detection at these levels would definitively falsify the hypothesis.

6.1 Theoretical Implications

If confirmed, a significant PBH component of dark matter would have profound implications for early universe physics and structure formation. It would provide direct evidence for enhanced density perturbations during inflation and would establish a concrete link between quantum fluctuations in the first 10^{-30} seconds and the large-scale structure we observe today.

Moreover, PBH-seeded structure formation may help resolve several outstanding puzzles in cosmology:

- The rapid formation of supermassive black holes at $z > 10$ (JWST observations)
- The origin of intermediate-mass black holes detected by LIGO/Virgo
- The “missing satellites” and “too-big-to-fail” problems in galaxy formation
- The self-similar, scale-invariant properties of the cosmic web [[Vazza & Feletti, 2020](#)]

6.2 Remaining Questions

Several important questions remain beyond the scope of this paper:

Complete dark matter budget: If PBHs constitute only 10–30% of apparent dark matter, what accounts for the remainder? Possibilities include Population III stellar remnants, baryonic dark matter in compact objects, emergent gravitational effects [Verlinde, 2017], or a combination of these. A complete model requires further theoretical development and observational constraints.

PBH clustering mechanism: While simulations predict PBH clustering, the detailed spatial distribution and its evolution over cosmic time require more sophisticated numerical modeling. This is crucial for accurate predictions of microlensing event rates and spatial distributions.

Formation mechanism: While we have adopted the standard inflationary formation scenario, alternative mechanisms (e.g., phase transitions, cosmic string collapse) may also produce PBHs. Distinguishing between formation mechanisms may require additional observational signatures beyond microlensing.

6.3 Observational Timeline

The next decade (2025–2035) will provide definitive tests of our model:

- **2025–2027:** LSST begins operations; early microlensing detections
- **2028–2033:** Roman GBTDS; high-cadence microlensing survey
- **2028–2030:** Euclid weak lensing survey; dark matter substructure constraints
- **2030–2035:** Combined analysis of Roman + LSST data; definitive test of PBH mass function

By 2035, we will know whether primordial black holes constitute a significant component of dark matter and whether they serve as the gravitational seeds for cosmic structure. Regardless of the outcome, these observations will provide crucial insights into the nature of dark matter and the formation of structure in the universe.

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